

Lecture 11

THE WORK OF A FORCE, THE PRINCIPLE OF WORK AND ENERGY & SYSTEMS OF PARTICLES

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THE WORK OF A FORCE, THE PRINCIPLE OF WORK AND ENERGY & SYSTEMS OF PARTICLES

Today's Objectives:

Students will be able to:

1. Calculate the work of a force.
2. Apply the principle of work and energy to a particle or system of particles.



In-Class Activities:

- Reading Quiz
- Applications
- Work of a Force
- Principle of Work and Energy
- Example Problem
- Concept Quiz
- Group Problem Solving
- Attention Quiz

READING QUIZ

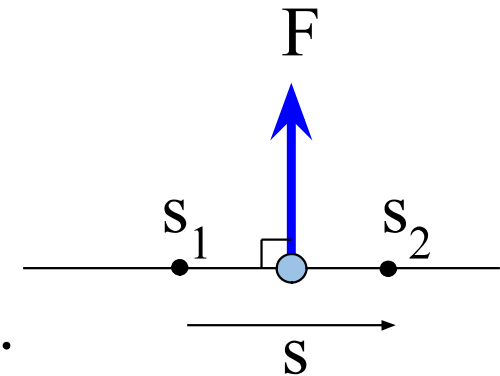
1. What is the work done by the force F ?

A) $F s$

B) $-F s$

C) Zero

D) None of the above.



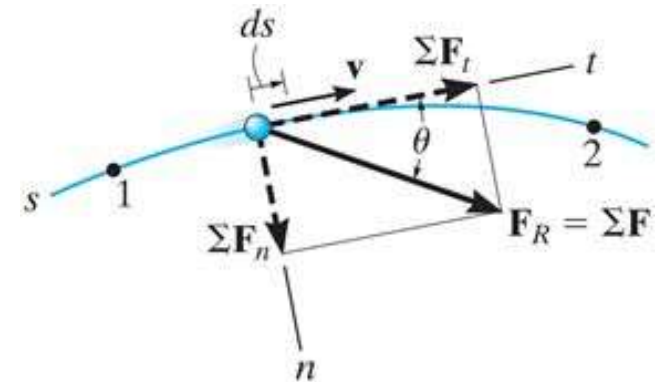
2. If a particle is moved from 1 to 2, the work done on the particle by the force, F_R will be

A) $\int_{s_1}^{s_2} \mathbf{F} \cdot d\mathbf{s}$

B) $-\int_{s_1}^{s_2} \Sigma F_t ds$

C) $\int_{s_1}^{s_2} \mathbf{F} \cdot d\mathbf{s}$

D) $-\int_{s_1}^{s_2} \Sigma F_n ds$



APPLICATIONS



A roller coaster makes use of gravitational forces to assist the cars in reaching high speeds in the “valleys” of the track.

How can we design the track (e.g., the height, h , and the radius of curvature, ρ) to control the forces experienced by the passengers?

APPLICATIONS (continued)



Crash barrels are often used along roadways in front of barriers for crash protection.

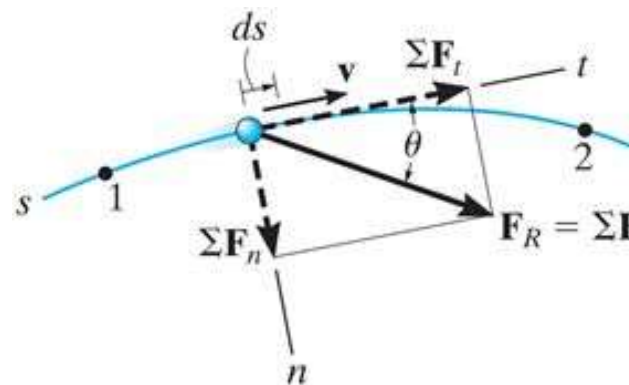
The barrels absorb the car's kinetic energy by deforming.

If we know the velocity of an oncoming car and the amount of energy that can be absorbed by each barrel, how can we design a crash cushion?

WORK AND ENERGY

Another equation for working kinetics problems involving particles can be derived by **integrating** the **equation of motion** ($\mathbf{F} = m\mathbf{a}$) with respect to **displacement**.

By substituting $a_t = v \, (dv/ds)$ into $F_t = ma_t$, the result is integrated to yield an equation known as the **principle of work and energy**.

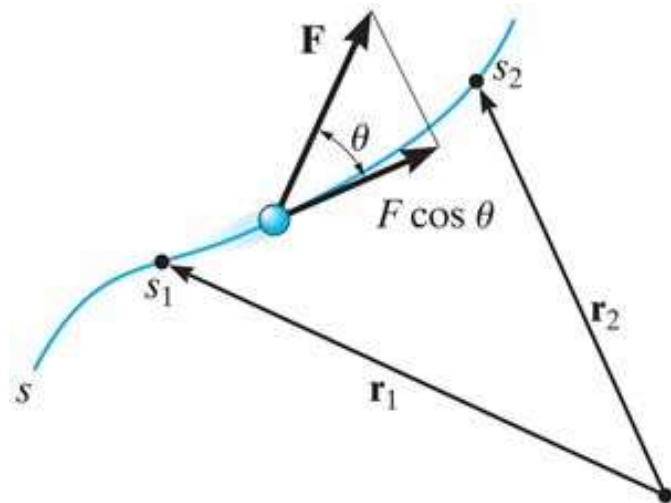


This principle is useful for solving problems that involve **force**, **velocity**, and **displacement**. It can also be used to explore the concept of **power**.

To use this principle, we must first understand how to calculate the **work of a force**.

WORK OF A FORCE (Section 14.1)

A force does work on a particle when the particle undergoes a displacement along the line of action of the force.



Work is defined as the **product** of **force** and **displacement components** acting in the **same direction**. So, if the angle between the force and displacement vector is θ , the increment of work dU done by the force is

$$dU = F \, ds \cos \theta$$

By using the definition of the **dot product** and integrating, the total work can be written as

$$U_{1-2} = \int_{r_1}^{r_2} \mathbf{F} \cdot d\mathbf{r}$$

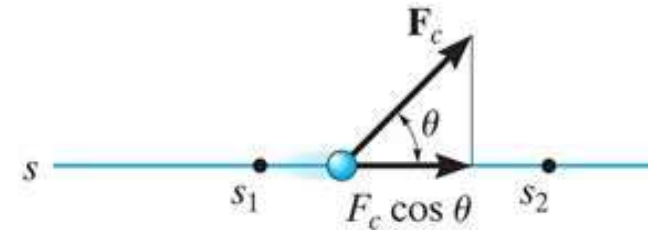
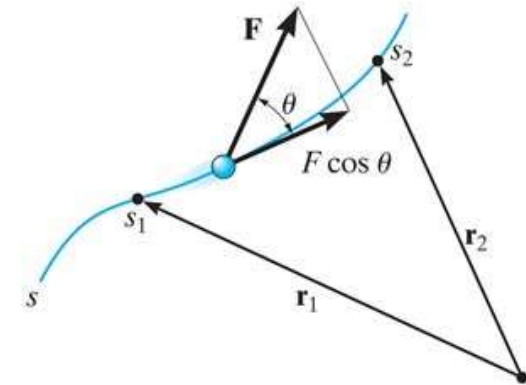
WORK OF A FORCE (continued)

If $\mathbf{F}$ is a function of position (a common case), this becomes

$$U_{1-2} = \int_{s_1}^{s_2} F \cos \theta \, ds$$

If both F and θ are constant ($F = F_c$), this equation further simplifies to

$$U_{1-2} = F_c \cos \theta (s_2 - s_1)$$



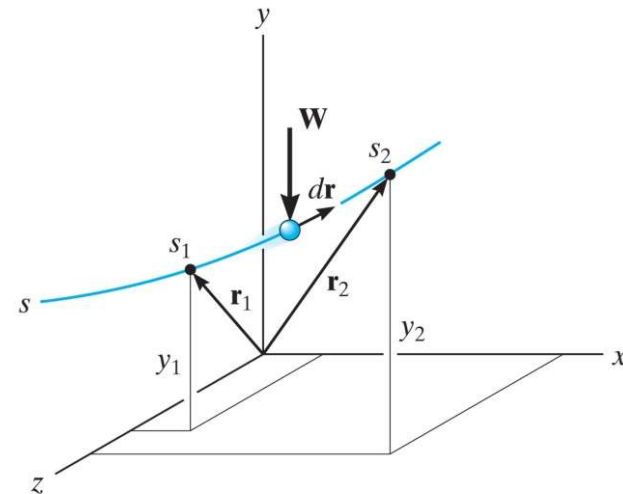
Work is **positive** if the force and the movement are in the **same direction**. If they are **opposing**, then the work is **negative**. If the force and the displacement directions are **perpendicular**, the work is **zero**.

WORK OF A WEIGHT

The work done by the gravitational force acting on a particle (or **weight of an object**) can be calculated by using

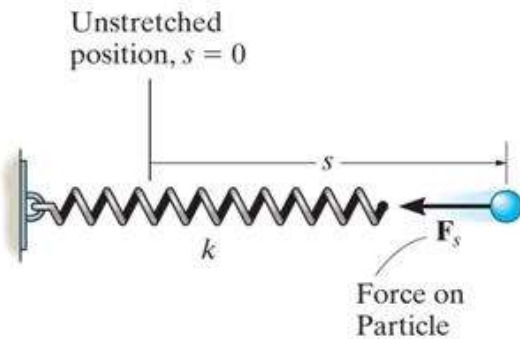
$$U_{1-2} = \int_{y_1}^{y_2} -W \, dy$$

$$U_{1-2} = -W (y_2 - y_1) = -W \Delta y$$



The work of a weight is the product of the magnitude of the particle's weight and its vertical displacement. If Δy is upward, the work is **negative** since the weight force always acts downward.

WORK OF A SPRING FORCE



When stretched, a **linear elastic spring** develops a force of magnitude $F_s = ks$, where k is the **spring stiffness** and s is the **displacement from the unstretched position**.

The work of the spring force moving from position s_1 to position s_2 is

$$U_{1-2} = \int_{s_1}^{s_2} F_s ds = \int_{s_1}^{s_2} k s ds = 0.5 k (s_2)^2 - 0.5 k (s_1)^2$$

If a particle is attached to the spring, the force F_s exerted **on the particle is opposite** to that exerted on the spring. Thus, the work done on the particle by the spring force will be **negative** or $U_{1-2} = -[0.5 k (s_2)^2 - 0.5 k (s_1)^2]$.

SPRING FORCES

It is important to note the following about spring forces.

1. The equations above are for **linear** springs only! Recall that a linear spring develops a force according to $F = ks$ (essentially the equation of a line).
2. The work of a spring is **not** just spring force times distance at some point, i.e., $(ks_i)(s_i)$. **Beware**, this is a trap that students often fall into!
3. Always **double check** the sign of the spring work after calculating it. It is positive work if the force on the object by the spring and the movement are in the same direction.

PRINCIPLE OF WORK AND ENERGY

(Section 14.2 & Section 14.3)

By integrating the equation of motion, $\sum F_t = ma_t = mv(dv/ds)$, the **principle of work and energy** can be written as

$$\sum U_{1-2} = 0.5 m (v_2)^2 - 0.5 m (v_1)^2 \quad \text{or} \quad T_1 + \sum U_{1-2} = T_2$$

$\sum U_{1-2}$ is the **work done by all the forces** acting on the particle as it moves from point 1 to point 2. Work can be either a **positive or negative** scalar.

T_1 and T_2 are the **kinetic energies** of the particle at the initial and final position, respectively. Thus, $T_1 = 0.5 m (v_1)^2$ and $T_2 = 0.5 m (v_2)^2$. The kinetic energy is always a **positive scalar** (velocity is squared!).

So, the **particle's initial kinetic energy plus the work done by all the forces acting on the particle as it moves from its initial to final position is equal to the particle's final kinetic energy.**

PRINCIPLE OF WORK AND ENERGY (continued)

Note that the principle of work and energy ($T_1 + \sum U_{1-2} = T_2$) is **not a vector equation**! Each term results in a scalar value.

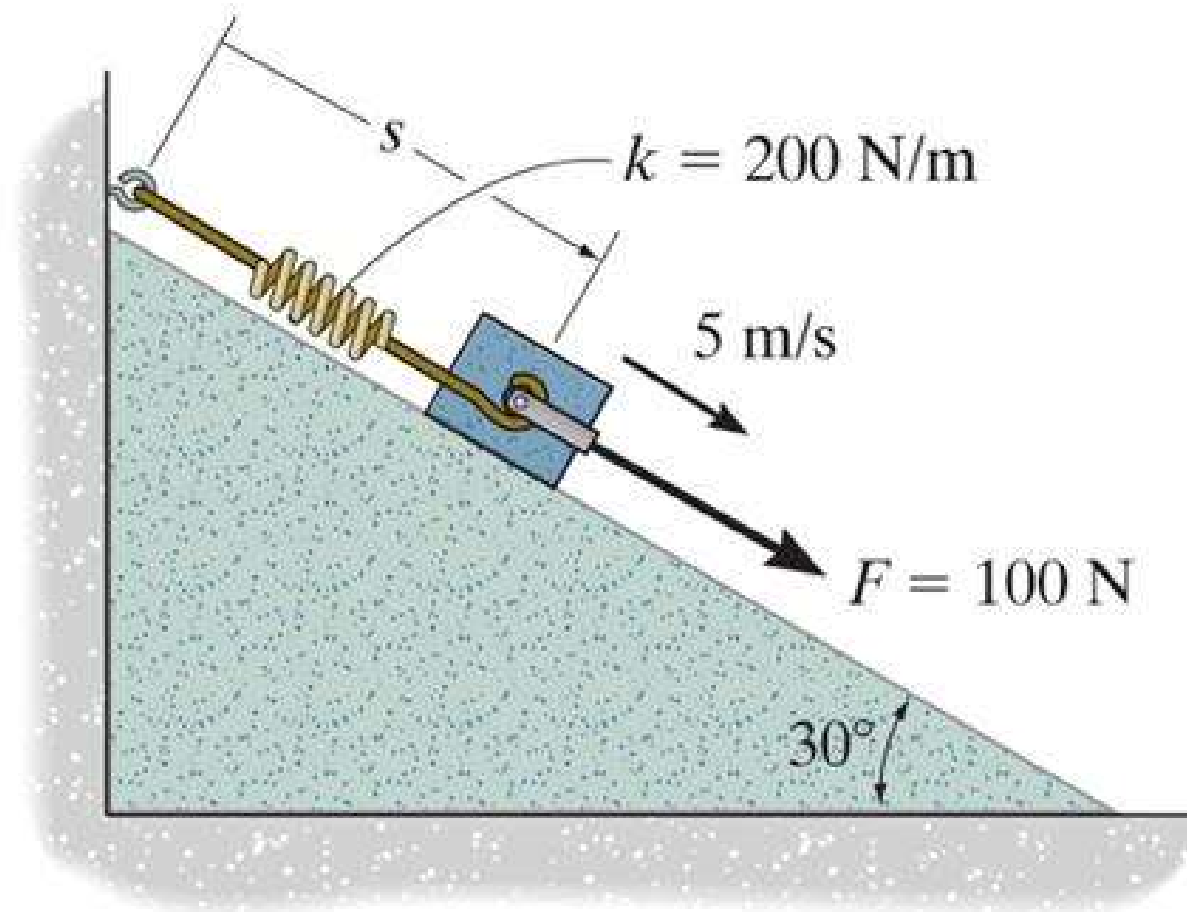
Both kinetic energy and work have the same units, that of energy! In the SI system, the unit for energy is called a **joule** (J), where $1 \text{ J} = 1 \text{ N}\cdot\text{m}$. In the FPS system, units are $\text{ft}\cdot\text{lb}$.

The principle of work and energy **cannot** be used, in general, to determine forces directed **normal** to the path, since these forces do no work.

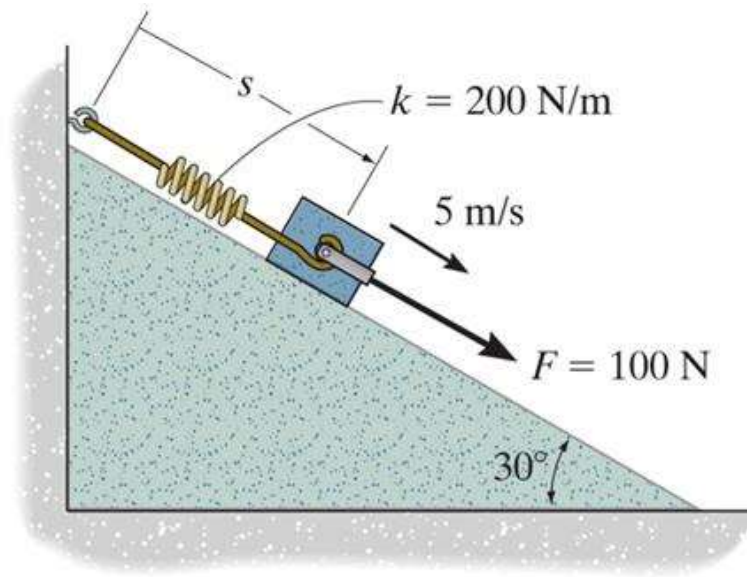
The principle of work and energy can also be applied to a **system of particles** by summing the kinetic energies of all particles in the system and the work due to all forces acting on the system.

EXAMPLE

When $s = 0.6$ m, the spring is unstretched and the 10-kg block has a speed of 5 m/s down the smooth plane. Determine the distance s when the block stops. **(Problem No. F14-5)**



EXAMPLE



Given: When $s = 0.6 \text{ m}$, the spring is not stretched or compressed, and the 10 kg block, which is subjected to a force of 100 N , has a speed of 5 m/s down the smooth plane.

Find: The distance s when the block stops.

Plan: Since this problem involves forces, velocity and displacement, apply the principle of work and energy to determine s .

EXAMPLE (continued)

Solution:

Apply the principle of work and energy between position 1 ($s_1 = 0.6$ m) and position 2 (s_2). Note that the normal force (N) does no work since it is always perpendicular to the displacement.

$$T_1 + \sum U_{1-2} = T_2$$

There is work done by three different forces;

- 1) work of the force $F = 100$ N;

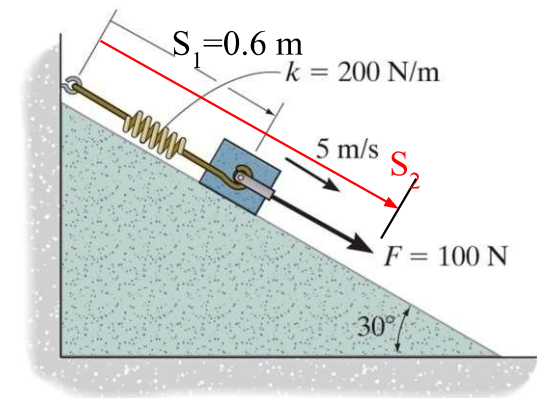
$$U_F = 100 (s_2 - s_1) = 100 (s_2 - 0.6)$$

- 2) work of the block weight;

$$U_W = 10 (9.81) (s_2 - s_1) \sin 30^\circ = 49.05 (s_2 - 0.6)$$

- 3) work of the spring force.

$$U_S = -0.5 (200) (s_2 - 0.6)^2 = -100 (s_2 - 0.6)^2$$



EXAMPLE (continued)

The work and energy equation will be

$$T_1 + \sum U_{1-2} = T_2$$

$$0.5 (10) 5^2 + 100(s_2 - 0.6) + 49.05(s_2 - 0.6) - 100(s_2 - 0.6)^2 = 0$$

$$\Rightarrow 125 + 149.05(s_2 - 0.6) - 100(s_2 - 0.6)^2 = 0$$

Solving for $(s_2 - 0.6)$,

$$(s_2 - 0.6) = \{-149.05 \pm (149.05^2 - 4 \times (-100) \times 125)^{0.5}\} / 2(-100)$$

Selecting the positive root, indicating a positive spring deflection,

$$(s_2 - 0.6) = 2.09 \text{ m}$$

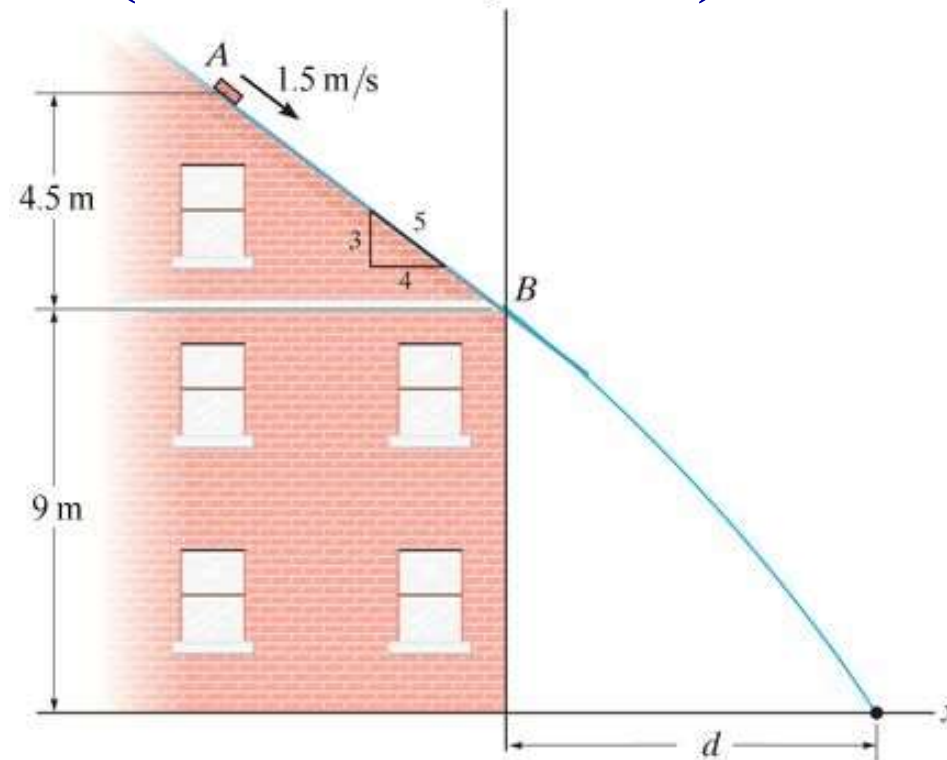
$$\text{Therefore, } s_2 = \underline{2.69 \text{ m}}$$

CONCEPT QUIZ

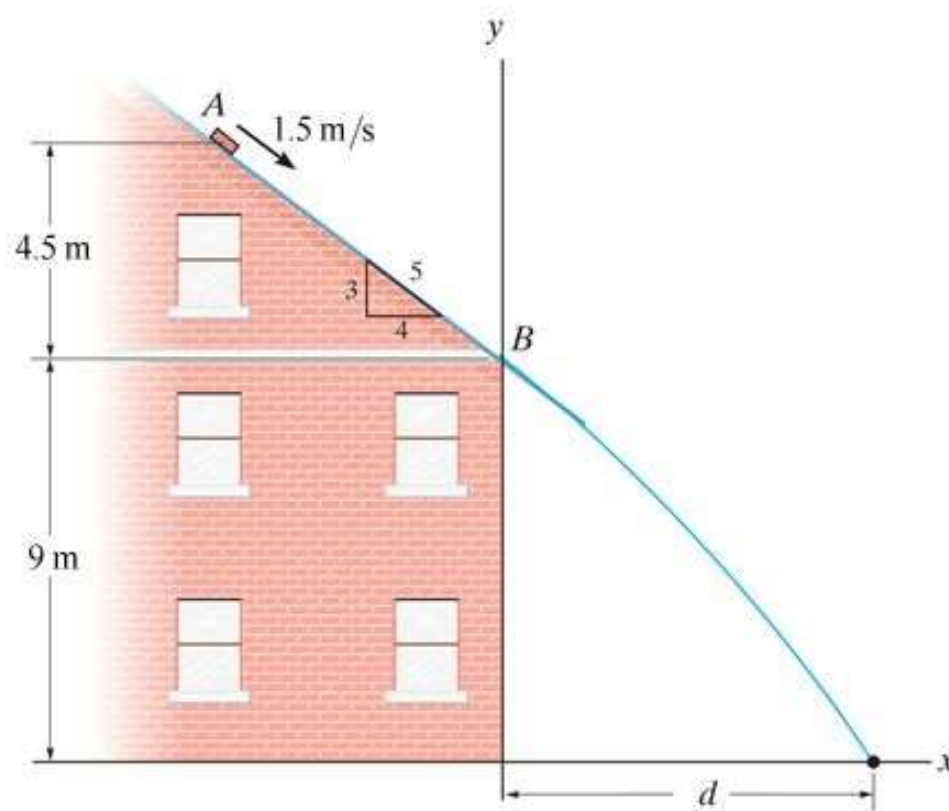
1. A spring with an unstretched length of 50 mm expands from a length of 20 mm to a length of 40 mm. The work done **on** the spring is _____ N·mm .
- A) $-[0.5 k(40 \text{ mm})^2 - 0.5 k(20 \text{ mm})^2]$ B) $0.5 k (20 \text{ mm})^2$
C) $-[0.5 k(30 \text{ mm})^2 - 0.5 k(10 \text{ mm})^2]$ D) $0.5 k(30 \text{ mm})^2 - 0.5 k(10 \text{ mm})^2$
2. If a spring force is $F = 5 s^3$ N/m and the spring is compressed by $s = 0.5$ m, the work done on a particle attached to the spring will be
- A) $0.625 \text{ N} \cdot \text{m}$ B) $-0.625 \text{ N} \cdot \text{m}$
C) $0.0781 \text{ N} \cdot \text{m}$ D) $-0.0781 \text{ N} \cdot \text{m}$

GROUP PROBLEM SOLVING

The 1-kg brick slides down a smooth roof, such that when it is at A it has a velocity of 1.5 m/s. Determine the speed of the brick just before it leaves the surface at B, the distance d from the wall to where it strikes the ground, and the speed at which it hits the ground. **(Problem No. 14-13)**



GROUP PROBLEM SOLVING



Given: The 1-kg brick slides down a smooth roof, with $v_A = 1.5 \text{ m/s}$.

Find: The speed at B, the distance d from the wall to where the brick strikes the ground, and its speed at C.

- Plan:**
- 1) Apply the principle of work and energy to the brick, and determine the speeds at B and C.
 - 2) Apply the kinematic relations in x and y -directions.

GROUP PROBLEM SOLVING I (continued)

Solution:

- 1) Apply the principle of work and energy

$$\sum T_A + \sum U_{A-B} = \sum T_B$$

$$\frac{1}{2} (1) (1.5)^2 + (1) (9.81)(4.5) = \frac{1}{2} (1) (v_B)^2$$

Solving for the unknown velocity yields $v_B = \underline{9.515 \text{ m/s}}$

Similarly, apply the work and energy principle between A and C

$$\sum T_A + \sum U_{A-C} = \sum T_C$$

$$\frac{1}{2} (1) (1.5)^2 + (1) (9.81)(13.5) = \frac{1}{2} (1) (v_C)^2$$

$$v_C = \underline{16.34 \text{ m/s}}$$

GROUP PROBLEM SOLVING I (continued)

2) Apply the kinematic relations in x and y-directions:

Equation for horizontal motion

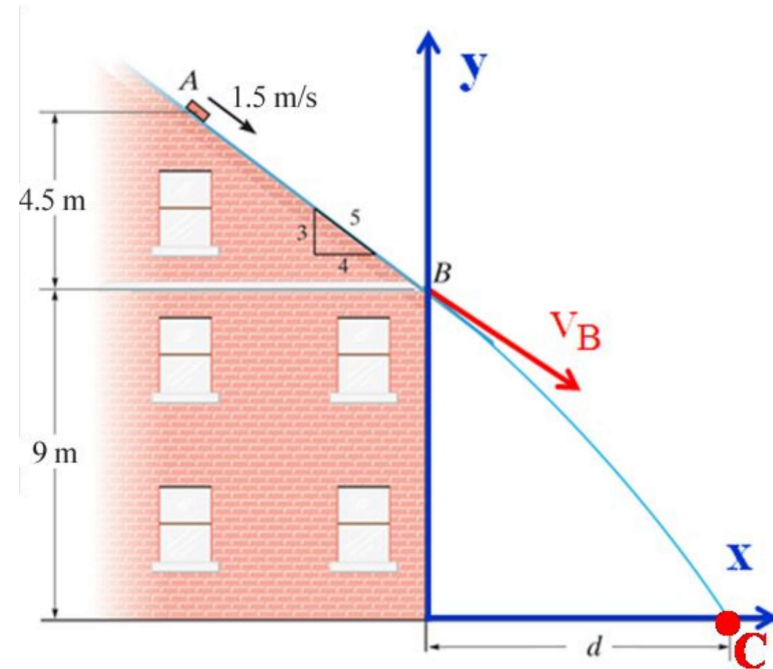
$$\begin{aligned} +\rightarrow x_C &= x_B + v_{Bx} t_{BC} \\ d &= 0 + 9.515 (4/5) t_{BC} \\ \Rightarrow d &= 7.612 t_{BC} \end{aligned}$$

Equation for vertical motion

$$\begin{aligned} +\uparrow y_C &= y_B + v_{By} t_{BC} - 0.5 g t_{BC}^2 \\ \Rightarrow -9 &= 0 + (-9.515)(3/5) t_{BC} - 0.5 (9.81) t_{BC}^2 \end{aligned}$$

Solving for the positive t_{BC} yields $t_{BC} = 0.8923 \text{ s}$.

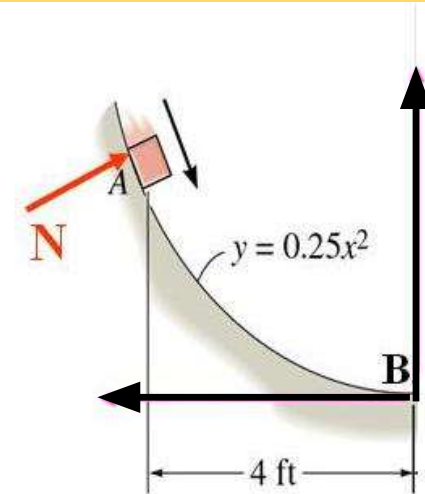
$$\Rightarrow d = 7.612 t_{BC} = 7.612 (0.8923) = \underline{6.79 \text{ m}}$$



ATTENTION QUIZ

1. What is the work done by the normal force N if a 2-kg box is moved from A to B ?

- A) $-78.5 \text{ N} \cdot \text{m}$ B) $0 \text{ N} \cdot \text{m}$
C) $78.5 \text{ N} \cdot \text{m}$ D) $157 \text{ N} \cdot \text{m}$



End of the Lecture
Let The Learning Continue